

# Geavanceerde computerarchitectuur: lab-3

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## 1 Learning objective

In this lab session we learn what warps are and how we can use threads efficiently if we take warps into account. We will also see what effect thread divergence has on the execution time of a CUDA program.

## 2 Exercise

In this exercise we will inverse the RGB values of a picture first by using the formula x = 256-x and later by using a different formula for certain R values.

### 2.1 Inverse

By calculating the inverse using CUDA we can experiment to see what the effect of warps are. We will use two different instances of thread blocks and threads per block. One being 16 blocks and 128 threads per block and the other being 14 blocks and 145 threads per block. This means one will have full warps and the other will have a partial warp. We expect this to have an effect on the computation speed because warps will have to iterate twice. If we compare the average computation speed as seen below we can conclude that the use of full warps is faster.

### 2.2 Thread divergence

When part of the computation requires extra complexity, like for example we add this for the R: x = (x % 25) $\cdot$ 10, x for (100, 200). This means other threads in the block will be finished faster then the ones with the complex calculation and will occupy the core with waiting. So we expect to see a drop in performance and this is the case if we look at the data. We want to avoid idle waiting. We could make sure that certain warps calculate the complex parts and others the simpler ones so the waiting is minimised or we could avoid branching in the first place.

## 3 Conclusions

When executing a program using the GPU we should take warps into account to make the execution time faster. If we need to add complexity to the calculations we first of all should avoid this if possible and if this is not the case we should try to combine the complex calculations in the same warps so that the idle waiting of threads that occupy a core are minimised.

| full warps | partial warps | thread divergence |
|---|---|---|
| 0,187168 | 0,196512 | 0,349504 |
| 0,12496 | 0,302144 | 0,422144 |
| 0,196224 | 0,28368 | 0,215808 |
| 0,22384 | 0,13488 | 0,254688 |
| 0,126624 | 0,123936 | 0,305248 |
| 0,176768 | 0,229408 | 0,47456 |
| 0,096832 | 0,217184 | 0,37376 |
| 0,167168 | 0,13616 | 0,297664 |
| 0,09584 | 0,253504 | 0,414848 |
| 0,180352 | 0,268544 | 0,38736 |
| 0,1575776 | 0,214595 | 0,349558 |

Table 1: executing speeds for inversing an image with CUDA in three different ways. Expressed in ms.

